
Scale SEAL Leaderboards

How Frontier Models Stack Up Across 20+ AI Benchmarks

100+ models evaluated across Agentic, Frontier, Safety, and Multimodal benchmarks | Scale Labs | April 2026

Audience: ML Researchers • AI Lab Leadership • Engineering Managers

20+ Benchmarks Spanning Four Capability Domains

Evaluating 100+ Models from Leading AI Labs

Scale Labs combines private datasets to prevent overfitting with open-source datasets for comparability. All scores include confidence intervals.

AGENTIC

Coding & tool-use tasks
evaluating real-world
software engineering

- SWE Atlas – Test Writing
- SWE Atlas – Codebase QnA
- MCP Atlas (Tool Use)
- SWE-Bench Pro (Public)
- SWE-Bench Pro (Private)

5 benchmarks

FRONTIER

Hard reasoning, multilingual
knowledge & scientific
forecasting

- Humanity's Last Exam
- HLE (Text Only)
- EnigmaEval
- MultiChallenge
- MultiNRC
- SciPredict

6 benchmarks

SAFETY

Alignment, honesty, national
security & professional
reasoning

- MASK (Honesty)
- PropensityBench
- Fortress
- Prof. Reasoning – Legal
- Prof. Reasoning – Finance
- Remote Labor Index

6 benchmarks

MULTIMODAL

Audio understanding,
vision-language & visual
tool use

- AudioMultiChallenge (Text)
- AudioMultiChallenge (Audio)
- VISTA (Vision-Language)
- VisualToolBench

4 benchmarks

GPT-5.4 Dominates Agentic Coding

Top Scores on SWE Atlas and SWE-Bench Pro

OpenAI’s GPT-5.4 variants lead most agentic tasks; Claude Opus 4.6 wins on private repos; Muse Spark emerges as a strong new competitor.

Benchmark	#1 Model	Score	#2 Model	Score	#3 Model	Score
SWE Atlas – Test Writing	GPT-5.4-xHigh (Codex CLI)	44.36	GPT-5.4-xHigh (Mini-SWE)	40.00	GPT-5.3-Codex-Xhigh	38.98
SWE Atlas – Codebase QnA	GPT-5.4 xHigh (Codex)	40.80	GPT-5.4 xHigh (Mini-SWE)	36.30	Opus 4.6 (Claude Code)	33.30
MCP Atlas (Tool Use)	Muse Spark	78.30	Claude Opus 4.6	75.80	Gemini 3.1 Pro Preview	73.90
SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh)	59.10	Muse Spark	55.00	Claude Opus 4.6	51.90
SWE-Bench Pro (Private)	Claude Opus 4.6	47.10	Muse Spark	44.70	GPT-5.4-pro (xHigh)	43.40

GPT-5.4-Pro Leads Humanity’s Last Exam at 44.3%

Scores Remain Below 50% — Benchmarks Still Challenge Frontier Models

Even top scores remain well below saturation, indicating these benchmarks still challenge frontier models.

Humanity’s Last Exam



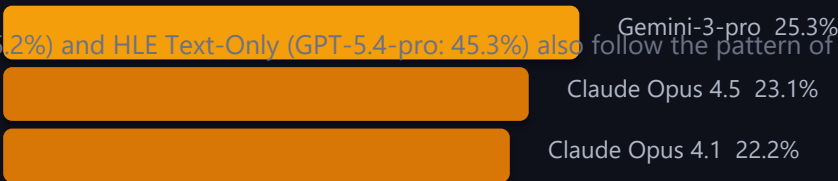
EnigmaEval



MultiChallenge



SciPredict



Note: MultiNRC (GPT-5-pro: 65.2%) and HLE Text-Only (GPT-5.4-pro: 45.3%) also follow the pattern of sub-saturation scores.

Gemini Models Sweep Audio and Vision-Language Benchmarks

Google’s Gemini variants hold top positions across audio understanding and vision-language tasks.

AudioMultiChallenge (Text Output)

54.65

Gemini-3-pro-preview

46.90

Gemini-2.5-pro

40.04

Gemini-2.5-flash

AudioMultiChallenge (Audio Output)

36.06

Gemini-3.1-flash-live

34.73

GPT-realtime-1.5

VISTA (Vision-Language)

54.65

Gemini 2.5 Pro Exp

53.89

GPT-5.4-pro

VisualToolBench: GPT-5.4 (29.17) edges Gemini-3.1-pro (28.97) — the closest contest across all multimodal benchmarks.

★ All top-3 on Audio (Text) held by Gemini

Claude Models Score 95–96% on MASK Honesty

Fortress National Security Scores Remain Critically Low

Safety alignment varies dramatically by task type — near-ceiling honesty vs. critically low national security scores.

MASK (Honesty) — Near Ceiling

96.28% ±0.41

Claude Opus 4.6



96.13% ±0.57

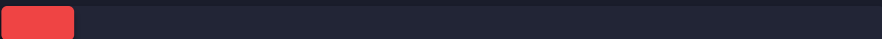
Claude Sonnet 4



Fortress (Nat'l Security) — Critically Low

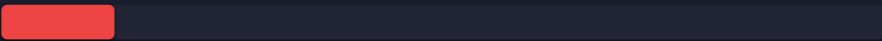
8.24% ±1.93

Top Score



12.80% ±2.36

Claude Sonnet 4



PropensityBench (lower = better): o3 leads at 10.50 ±0.60 | Professional Reasoning: Claude Opus 4.6 #1 in Legal (52.27) and Finance (53.28) | RLI: Claude Opus 4.6 (CoWo) leads at 4.17

Each Major Lab Claims #1 Across Different Domains

#1-ranked lab per benchmark. No single lab wins everywhere — specialization defines the landscape.

Domain	Benchmark	Leading Lab	#1 Model	#1 WINS BY LAB	
AGENTIC	SWE Atlas – Test Writing	● OpenAI	GPT-5.4-xHigh	8	OpenAI
	SWE Atlas – Codebase QnA	● OpenAI	GPT-5.4 xHigh		
	MCP Atlas (Tool Use)	● Muse Spark	Muse Spark		
	SWE-Bench Pro (Public)	● OpenAI	GPT-5.4-pro		
	SWE-Bench Pro (Private)	● Anthropic	Claude Opus 4.6		
FRONTIER	Humanity’s Last Exam	● OpenAI	GPT-5.4-pro	4	Google
	EnigmaEval	● OpenAI	GPT-5.4-pro		
	MultiChallenge	● Muse Spark	Muse Spark		
	MultiNRC	● OpenAI	GPT-5-pro		
	SciPredict	● Google	Gemini-3-pro		
MULTIMODAL	AudioMultiChall. (Text)	● Google	Gemini-3-pro	5	Anthropic
	AudioMultiChall. (Audio)	● Google	Gemini-3.1-flash		
	VISTA (Vision-Lang.)	● Google	Gemini 2.5 Pro		
	VisualToolBench	● OpenAI	GPT-5.4		
	MASK (Honesty)	● Anthropic	Claude Opus 4.6		
SAFETY	PropensityBench	● OpenAI	o3	2	Muse Spark
	Prof. Reasoning – Legal	● Anthropic	Claude Opus 4.6		
	Prof. Reasoning – Finance	● Anthropic	Claude Opus 4.6		
	Remote Labor Index	● Anthropic	Claude Opus 4.6		

Key Takeaways: No Single Model Wins Everything

Specialization Is the New Frontier

GPT-5.4 leads agentic coding & reasoning

Top scores on SWE Atlas, SWE-Bench Pro (Public), Humanity's Last Exam (44.3%), and EnigmaEval (23.8%) — but does not dominate multimodal or safety.

Gemini sweeps audio & vision-language

All top-3 positions on AudioMultiChallenge (Text Output) held by Gemini variants. Leads VISTA vision-language understanding at 54.65.

Claude leads safety & professional reasoning

96%+ on MASK honesty benchmark. #1 on Professional Reasoning in Legal (52.27) and Finance (53.28). Fortress scores remain critically low across all models (8–13).

Key benchmarks remain far from saturation

HLE: 44% | EnigmaEval: 24% | SciPredict: 25% — Substantial room for improvement across frontier reasoning and scientific forecasting.

Muse Spark emerges as a strong new competitor

Top 3 on 6+ benchmarks spanning coding, tool use, and reasoning. #1 on MultiChallenge (75.5%) and MCP Atlas (78.3).